

Checkpoint 2 Report: Pneumonia Detection from Chest X-Rays

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Project Goal

The goal is to build a binary image classifier that distinguishes **NORMAL** and **PNEUMONIA** chest X-ray images.

Dataset: Kaggle - Chest X-Ray Images (Pneumonia) by Paul Mooney.

Project Resources

GitHub Repository

https://github.com/loonaarc/Adv_AI_Project_Pneumonia/tree/main

Notebook on GitHub

https://github.com/loonaarc/Adv_AI_Project_Pneumonia/blob/main/pneumonia_detection_checkpoint2.ipynb

Open Notebook in Google Colab

https://colab.research.google.com/github/loonaarc/Adv_AI_Project_Pneumonia/blob/main/pneumonia_detection_checkpoint2.ipynb

The notebook is designed to run both locally in VS Code and in Google Colab. It downloads the dataset through the Kaggle API, so a Kaggle API token is required. In Colab, the recommended setup is to add **KAGGLE_API_TOKEN** in the Colab Secrets panel.

Data Understanding

The original Kaggle dataset contains separate **train**, **val**, and **test** folders. The provided validation split is very small:

Original split	NORMAL	PNEUMONIA	Total
train	1341	3875	5216
val	8	8	16
test	234	390	624

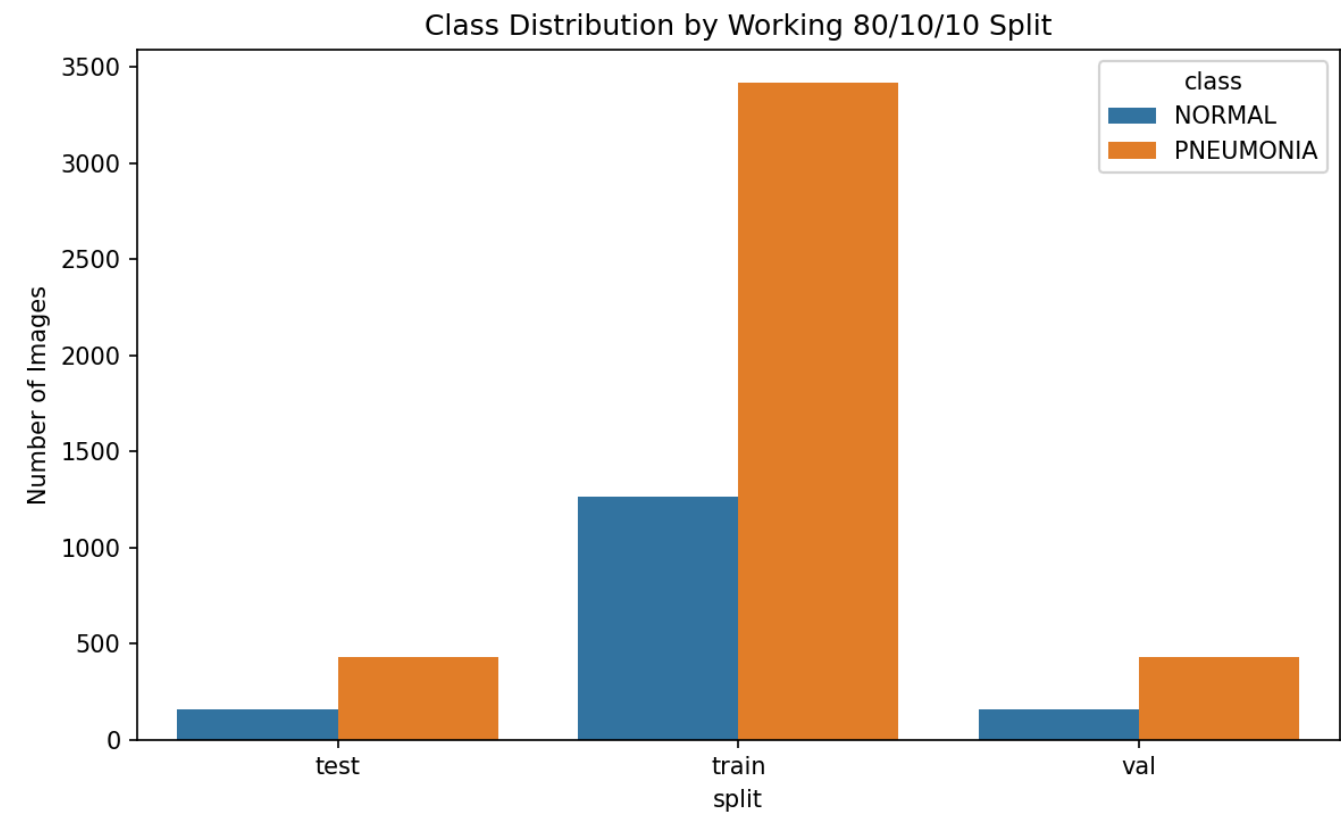
Because 16 validation images are not enough for reliable model selection, the notebook creates a new reproducible stratified 80/10/10 split across all downloaded images.

Working split	NORMAL	PNEUMONIA	Total
train	1266	3418	4684
val	159	427	586

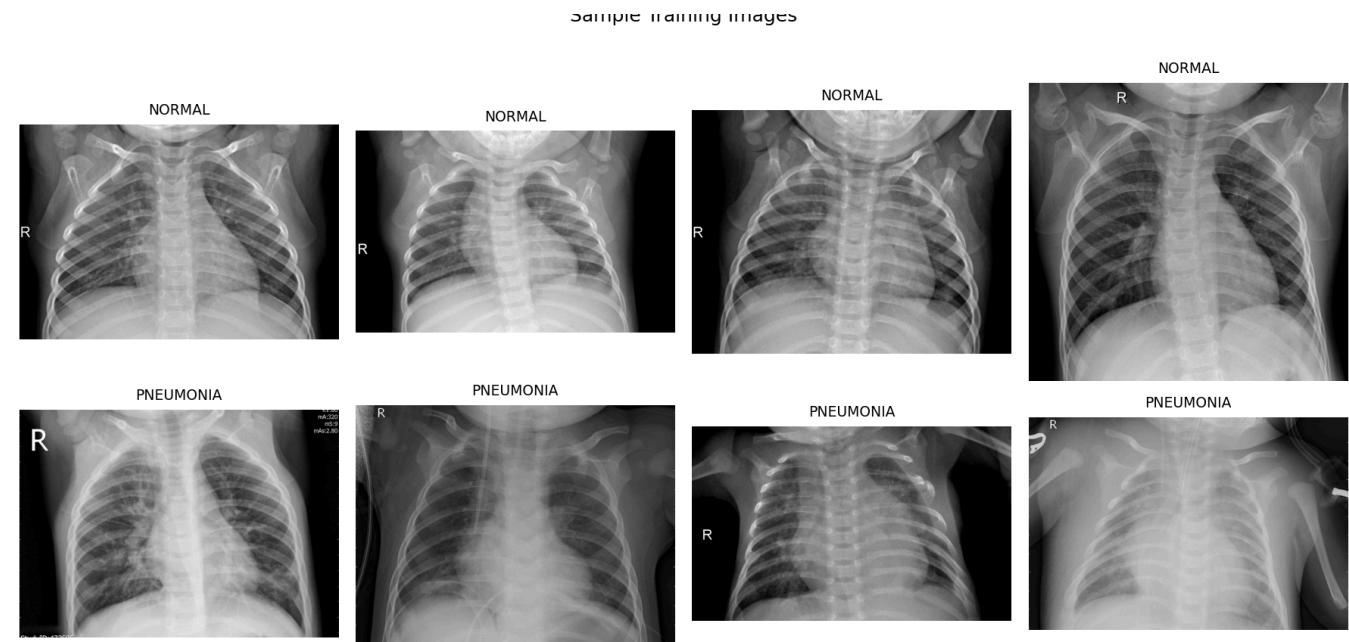
Working split	NORMAL	PNEUMONIA	Total
test	158	428	586

The split is saved to `outputs/results/dataset_split_80_10_10.csv`.

Class distribution for the working split:



Example training images:



Data Preprocessing

Images are loaded from the split manifest and processed as follows:

- decode image files as grayscale
- resize to 224 x 224
- convert grayscale images to 3 channels for CNN compatibility
- normalize pixel values to [0, 1]
- apply data augmentation only to the training split

Training augmentation includes small rotations, zoom, horizontal flip, and contrast variation.

Baseline Model

The baseline is a small convolutional neural network:

- 3 convolution + max pooling blocks
- dropout for regularization
- dense hidden layer
- sigmoid output for binary classification

The model uses Adam, binary cross-entropy loss, and reports accuracy, precision, and recall.

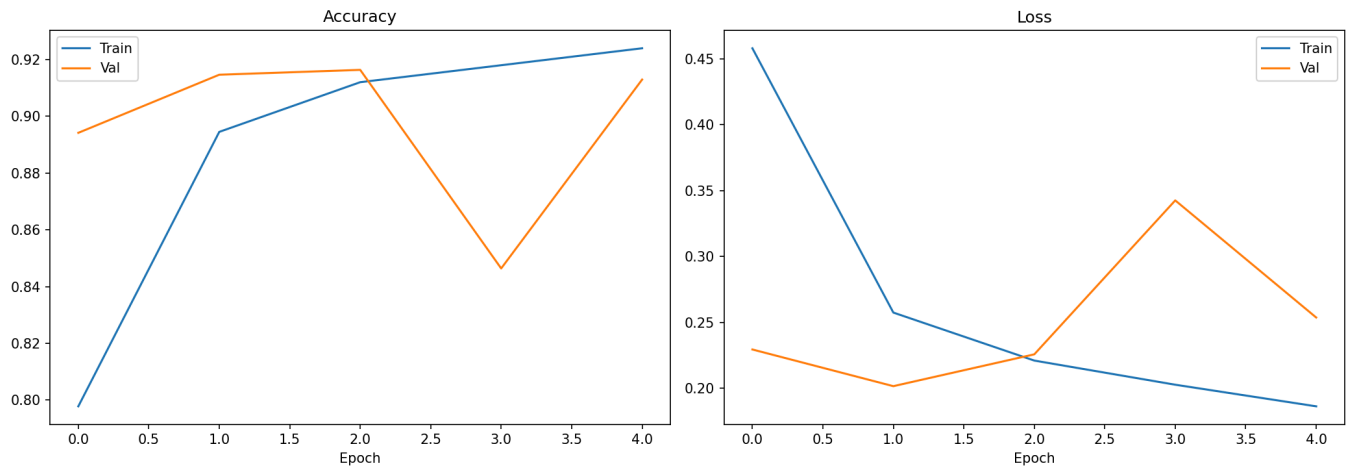
Baseline Results

The baseline CNN was trained for 5 epochs with early stopping monitoring validation loss. The validation curves are not fully smooth: training loss decreases steadily, while validation loss improves at first and then becomes unstable, including one epoch with a clear validation accuracy drop. This suggests that the baseline model is useful as a first reference, but not yet robust enough for final conclusions.

This instability is a key checkpoint finding. It may be caused by the class imbalance, the single random split, the small CNN architecture, or difficult individual X-ray images. Therefore, the test metrics below are treated as baseline evidence, not as final proof that the model is reliable.

Metric	Test result
Loss	0.1909
Accuracy	0.9283
Precision	0.9488
Recall	0.9533

Baseline training curves:



The unstable validation curve in this baseline run shows a limitation of relying on one validation split. Simpler ways to reduce this issue would be to repeat training with different random seeds, apply class weighting, or tune the classification threshold. Stratified k-fold cross-validation could also help if more compute is available.

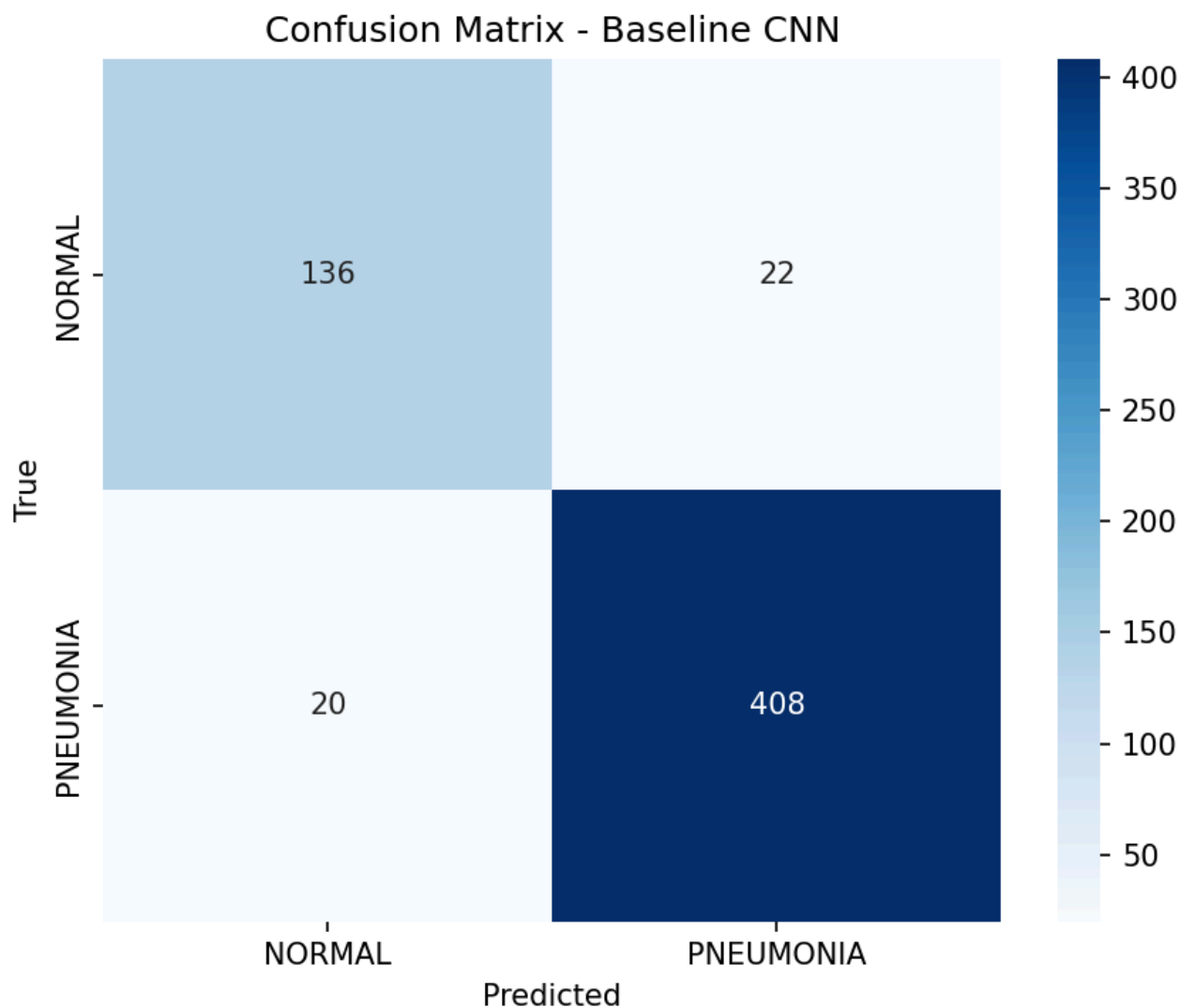
Detailed test classification report:

Class	Precision	Recall	F1-score	Support
NORMAL	0.8718	0.8608	0.8662	158
PNEUMONIA	0.9488	0.9533	0.9510	428
Macro avg	0.9103	0.9070	0.9086	586
Weighted avg	0.9281	0.9283	0.9282	586

Confusion matrix on the test split:

True class	Predicted NORMAL	Predicted PNEUMONIA
NORMAL	136	22
PNEUMONIA	20	408

Confusion matrix figure:



Generated result files:

- `outputs/results/baseline_metrics.txt`
- `outputs/results/classification_report.txt`
- `outputs/figures/training_curves.png`
- `outputs/figures/confusion_matrix.png`

Initial Observations

- The dataset is strongly imbalanced toward pneumonia cases.
- The original validation split is too small, so an 80/10/10 stratified split is used for a more stable validation signal.
- The baseline CNN reaches promising test performance, especially for pneumonia recall, but the validation curves are unstable.
- The unstable validation curve shows that the single validation split should be interpreted cautiously; class weighting, threshold tuning, or repeated runs could make the estimate more reliable.
- The confusion matrix shows 20 false negatives for pneumonia and 22 false positives for pneumonia. For a medical screening task, false negatives are especially important and should be reduced further.
- The baseline CNN provides a first performance reference, but it should not be treated as a final reliable medical model.